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Intro to Finite Element Method (CIV E – 665)
Assignment No. 4



1 Question 1

Consider the following differential equation:

$$-\frac{d}{dx} \left(\frac{1}{5+x} \frac{du}{dx} \right) = x \quad 0 \leq x \leq 1$$

Subject to:

$$u(0) = 0$$
$$\frac{1}{5+x} \frac{du}{dx} \Big|_1 = 1$$

1.1 I. Find the analytical solution

Multiply by -1 both sides,

$$\frac{d}{dx} \left(\frac{1}{5+x} \frac{du}{dx} \right) = -x \quad (1)$$

Integrate both sides,

$$\int \frac{d}{dx} \left(\frac{1}{5+x} \frac{du}{dx} \right) dx = \int -x dx \quad (2)$$

This yields,

$$\frac{1}{5+x} \frac{du}{dx} = -\frac{1}{2}x^2 + C_1 \quad (3)$$

Apply $\frac{1}{5+x} \frac{du}{dx} \Big|_{x=1} = 1$ to Equation 3

Substitute $x = 1$ into our equation from Step 1,

$$-\frac{1}{2}(1)^2 + C_1 = 1 \implies C_1 = 1.5 \quad (4)$$

Substitute $x=1$ into Equation 3 and re-arrange it,

$$\frac{du}{dx} = (5+x) \left(-\frac{1}{2}x^2 + \frac{3}{2} \right) = -\frac{1}{2}x^3 - \frac{5}{2}x^2 + \frac{3}{2}x + \frac{15}{2} \quad (5)$$

For Integrate both sides with respect to x to find $u(x)$:

$$\begin{aligned}
u(x) &= \int \left(-\frac{1}{2}x^3 - \frac{5}{2}x^2 + \frac{3}{2}x + \frac{15}{2} \right) dx \\
&= -\frac{1}{2} \left(\frac{x^4}{4} \right) - \frac{5}{2} \left(\frac{x^3}{3} \right) + \frac{3}{2} \left(\frac{x^2}{2} \right) + \frac{15}{2}x + C_2 \\
&= -\frac{1}{8}x^4 - \frac{5}{6}x^3 + \frac{3}{4}x^2 + \frac{15}{2}x + C_2
\end{aligned} \tag{6}$$

Apply $u(0) = 0$ to find C_2 , substitute $x = 0$ into the expression for $u(x)$:

$$0 = -\frac{1}{8}(0)^4 - \frac{5}{6}(0)^3 + \frac{3}{4}(0)^2 + \frac{15}{2}(0) + C_2 \implies C_2 = 0 \tag{7}$$

Therefore, obtain the analytical solution

$$u(x) = -\frac{1}{8}x^4 - \frac{5}{6}x^3 + \frac{3}{4}x^2 + \frac{15}{2}x \tag{8}$$

1.2 II. Find the approximate solution using an $n = 2$ Ritz/Galerkin approximation

$$u(x) \approx c_0 + c_1x + c_2x^2 \tag{9}$$

Apply B.C $u(0)=0$, obtain $c_0 = 0$. And then $u(x)$ be simplified as:

$$u(x) = c_1x + c_2x^2 \tag{10}$$

Based on Ritz/Galerkin methon, build a weight function $w(x)$, and multiply the differential equation by $w(x)$ and integrate over the domain $[0,1]$,

$$\int_0^1 w \left[-\frac{d}{dx} \left(\frac{1}{5+x} \frac{du}{dx} \right) - x \right] dx = 0 \tag{11}$$

Integrate by parts,

$$\int_0^1 \frac{1}{5+x} \frac{dw}{dx} \frac{du}{dx} dx - \left[w \frac{1}{5+x} \frac{du}{dx} \right]_0^1 = \int_0^1 wx dx \tag{12}$$

Apply B. C. $\frac{1}{5+x} \frac{du}{dx} = 1$ into Equation 12. Since $u(0)=0$, have to set $w(0)=0$ to comply $u(x)$, so the weak form equation can be obtained,

$$\int_0^1 \frac{1}{5+x} w' u' dx = \int_0^1 wx dx + w(1) \tag{13}$$

Per the form of Equation 10, set ϕ as follows:

$$\phi_1(x) = x \implies \phi'_1 = 1$$

$$\phi_2(x) = x^2 \implies \phi'_2 = 2x$$

Substitute $u = c_1\phi_1 + c_2\phi_2$ into the weak form for each weight function $w = \phi_i$. This creates a system of linear equations: $[K]\{c\} = \{F\}$.

Construct stiffness matrix K , $K_{ij} = \int_0^1 \frac{1}{5+x} \phi_i' \phi_j' dx$

K_{11} (using $w = x, u = x$):

$$K_{11} = \int_0^1 \frac{1}{5+x} (1)(1) dx = [\ln(5+x)]_0^1 = \ln(6) - \ln(5) = \ln(1.2) \quad (14)$$

$K_{12} = K_{21}$ (using $w = x, u = x^2$):

$$K_{12} = \int_0^1 \frac{1}{5+x} (1)(2x) dx = 2 \int_0^1 \left(1 - \frac{5}{5+x}\right) dx \quad (15)$$

$$= 2 [x - 5 \ln(5+x)]_0^1 = 2(1 - 5 \ln(1.2)) \quad (16)$$

K_{22} (using $w = x^2, u = x^2$):

$$K_{22} = \int_0^1 \frac{1}{5+x} (2x)(2x) dx = 4 \int_0^1 \frac{x^2}{5+x} dx = 4(-4.5 + 25 \ln(1.2)) \quad (17)$$

Construct load vector items $\{F\}$: $F_i = \int_0^1 \phi_i x dx + \phi_i(1)$

F_1 (using $w = x$):

$$F_1 = \int_0^1 (x)(x) dx + (1) = \left[\frac{x^3}{3}\right]_0^1 + 1 = \frac{1}{3} + 1 = \frac{4}{3} \quad (18)$$

F_2 (using $w = x^2$):

$$F_2 = \int_0^1 (x^2)(x) dx + (1)^2 = \left[\frac{x^4}{4}\right]_0^1 + 1 = \frac{1}{4} + 1 = \frac{5}{4} \quad (19)$$

Therefore, the linear system is:

$$\begin{bmatrix} \ln\left(\frac{6}{5}\right) & 2\left(1 - 5 \ln\left(\frac{6}{5}\right)\right) \\ 2\left(1 - 5 \ln\left(\frac{6}{5}\right)\right) & 4\left(-4.5 + 25 \ln\left(\frac{6}{5}\right)\right) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} \frac{4}{3} \\ \frac{5}{4} \end{bmatrix} \quad (20)$$

Use Matlab to solve this system get c_1 and c_2 , the code as follows:

```
K11 = log(6/5);
K12 = 2*(1 - 5*log(6/5));
K22 = 4*(-4.5 + 25*log(6/5));
K = [K11 K12; K12 K22];
F = [4/3; 5/4];

c = K\F;
c1 = c(1);
```

```
c2 = c(2);

fprintf('c1 = %.6f, c2 = %.6f\n', c1, c2);
```

Output:

```
c1 = 7.996964, c2 = -0.705297
```

Thus, the approximate solution is:

$$u(x) \approx 7.9969638x - 0.7052971x^2 \quad (21)$$

Plot the Exact and the Approximate Solution using the Ritz/Galerkin Method as shown in Figure 1:

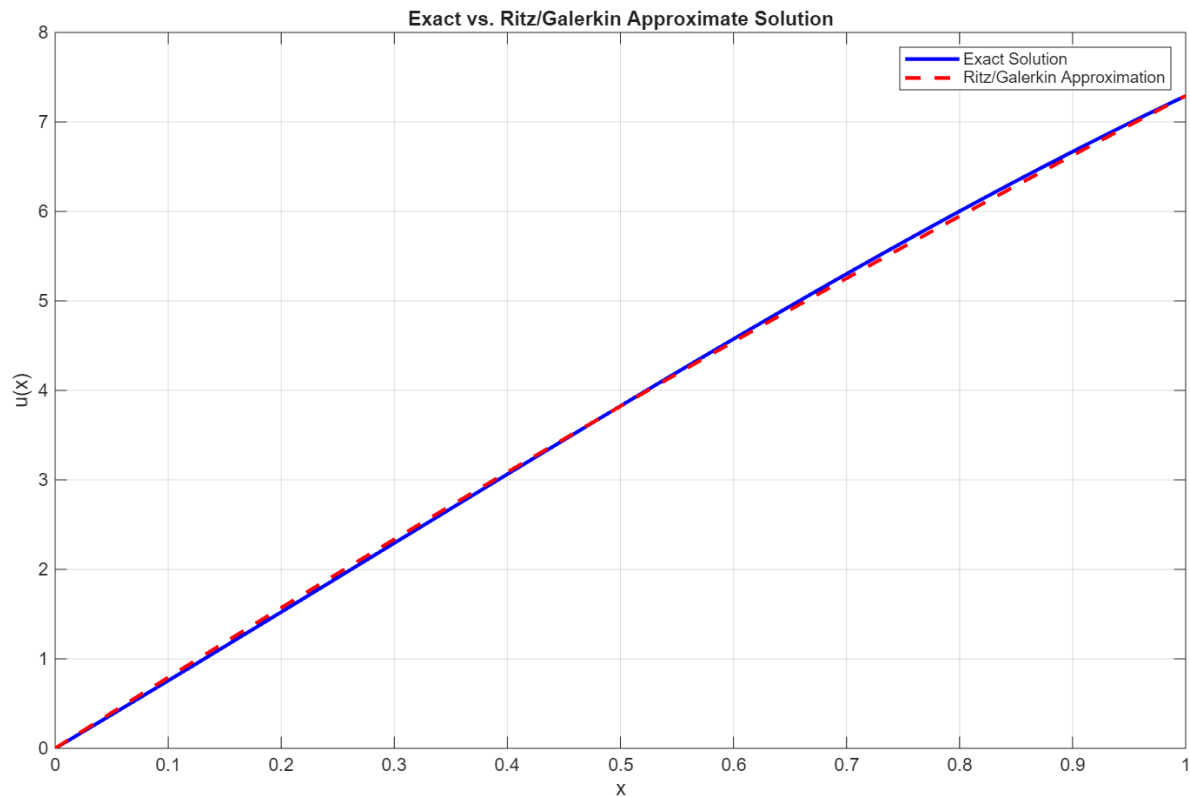


Figure 1: The comparison between exact and Ritz/Galerkin solutions

```
% Exact solution
x = linspace(0, 1, 400);
u_exact = -(1/8)*x.^4 - (5/6)*x.^3 + (3/4)*x.^2 + (15/2)*x;
% Ritz/Galerkin approximate solution
c1 = 7.9969638;
c2 = -0.7052971;
u_approx = c1*x + c2*x.^2;

plot(x, u_exact, 'b-', 'LineWidth', 2);
hold on;
```

```

plot(x, u_approx, 'r--', 'LineWidth', 2);
legend('Exact Solution', 'Ritz/Galerkin Approximation');
xlabel('x');
ylabel('u(x)');
title('Exact vs. Ritz/Galerkin Approximate Solution');
grid on;
hold off;

```

1.3 III. Find the approximate solution using FEM with two linear elements

1.3.1 Discretization

Divide the domain $0 \leq x \leq 1$ into two linear elements of equal length.

Nodes: 3 Nodes ($x_1 = 0, x_2 = 0.5, x_3 = 1$)

Elements:

- Element 1 (e_1): Nodes 1 and 2 ($0 \leq x \leq 0.5$), Length $h_1 = 0.5$
- Element 2 (e_2): Nodes 2 and 3 ($0.5 \leq x \leq 1$), Length $h_2 = 0.5$

Global Displacement Vector: $\{U\} = [U_1, U_2, U_3]^T$

1.3.2 Derive element wise

Similarly to Equation 13, we can get the weak form by integration by parts.

$$\int_0^1 \left(\frac{1}{5+x} \right) \frac{dw}{dx} \frac{du}{dx} dx - \left[w \left(\frac{1}{5+x} \frac{du}{dx} \right) \right]_0^1 = \int_0^1 wx dx \quad (22)$$

Rearranging to isolate the integral terms:

$$\int_0^1 \frac{1}{5+x} \frac{dw}{dx} \frac{du}{dx} dx = \int_0^1 wx dx + [w \cdot \text{Flux}]_0^1 \quad (23)$$

For a generic element e with nodes 1 and 2, we use linear shape functions ψ_1^e and ψ_2^e .

Shape Functions:

$$\psi_1^e(x) = \frac{x_2^e - x}{h_e}, \quad \psi_2^e(x) = \frac{x - x_1^e}{h_e} \quad (24)$$

Derivatives:

$$\frac{d\psi_1^e}{dx} = -\frac{1}{h_e}, \quad \frac{d\psi_2^e}{dx} = \frac{1}{h_e} \quad (25)$$

1.3.2.1 Element Stiffness Matrix $[K^e]$

The stiffness matrix components are defined by the integral on the LHS of the weak form:

$$K_{ij}^e = \int_{x_1^e}^{x_2^e} \frac{1}{5+x} \frac{d\psi_i^e}{dx} \frac{d\psi_j^e}{dx} dx \quad (26)$$

Using $h_e = 0.5$, the derivatives are ± 2 . Thus, $\frac{d\psi_i}{dx} \frac{d\psi_j}{dx}$ is either 4 (if $i = j$) or -4 (if $i \neq j$).

$$K_{ij}^e = \pm 4 \int_{x_1^e}^{x_2^e} \frac{1}{5+x} dx = \pm 4 \left[\ln(5+x) \right]_{x_1^e}^{x_2^e} \quad (27)$$

For element 1 (0 to 0.5):

$$k_{val}^1 = 4(\ln(5.5) - \ln(5)) = 4 \ln(1.1) \quad (28)$$

$$[K^1] = \begin{bmatrix} k_{val}^1 & -k_{val}^1 \\ -k_{val}^1 & k_{val}^1 \end{bmatrix} \quad (29)$$

For element 2 (0.5 to 1):

$$k_{val}^2 = 4(\ln(6) - \ln(5.5)) = 4 \ln\left(\frac{12}{11}\right) \quad (30)$$

$$[K^2] = \begin{bmatrix} k_{val}^2 & -k_{val}^2 \\ -k_{val}^2 & k_{val}^2 \end{bmatrix} \quad (31)$$

1.3.2.2 Element Force Vector $\{f^e\}$

The force vector comes from the source term integral:

$$f_i^e = \int_{x_1^e}^{x_2^e} x \cdot \psi_i^e(x) dx \quad (32)$$

Element 1 (e^1):

$$f_1^1 = \int_0^{0.5} x \frac{0.5-x}{0.5} dx = \frac{1}{24} \quad (33)$$

$$f_2^1 = \int_0^{0.5} x \frac{x}{0.5} dx = \frac{1}{12} \quad (34)$$

Element 2 (e^2):

$$f_1^2 = \int_{0.5}^1 x \frac{1-x}{0.5} dx = \frac{1}{6} \quad (35)$$

$$f_2^2 = \int_{0.5}^1 x \frac{x-0.5}{0.5} dx = \frac{5}{24} \quad (36)$$

1.3.3 Global Assembly

We assemble the element matrices into the global system $[K]\{U\} = \{F\}$ by summing contributions at shared nodes (specifically Node 2, which is shared by e^1 and e^2).

$$\begin{bmatrix} K_{11}^1 & K_{12}^1 & 0 \\ K_{21}^1 & K_{22}^1 + K_{11}^2 & K_{12}^2 \\ 0 & K_{21}^2 & K_{22}^2 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix} = \begin{bmatrix} f_1^1 \\ f_2^1 + f_1^2 \\ f_2^2 \end{bmatrix} \quad (37)$$

Substituting the calculated values:

$$\begin{bmatrix} k_{val}^1 & -k_{val}^1 & 0 \\ -k_{val}^1 & k_{val}^1 + k_{val}^2 & -k_{val}^2 \\ 0 & -k_{val}^2 & k_{val}^2 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{24} \\ \frac{1}{12} + \frac{1}{6} \\ \frac{5}{24} \end{bmatrix} \quad (38)$$

1.3.4 Impose B.C

For essential B.C, from the problem statement $u(0) = 0$. We modify the first row/column to enforce $U_1 = 0$. We can eliminate the first equation or explicitly set the diagonal to 1 and the RHS to 0.

For natural B.C (at $x = 1$): The problem specifies $\frac{1}{5+x} \frac{du}{dx} \Big|_{x=1} = 1$. The boundary term from the weak form is $[w \cdot \text{Flux}]_0^1$. At node 3 ($x = 1$), the weight function $\psi_2^2(1) = 1$. We add the flux value (+1) directly to the global force vector at Node 3 (F_3).

Reduced System (solving for U_2, U_3):

$$\begin{bmatrix} k_{val}^1 + k_{val}^2 & -k_{val}^2 \\ -k_{val}^2 & k_{val}^2 \end{bmatrix} \begin{bmatrix} U_2 \\ U_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{4} \\ \frac{5}{24} + 1 \end{bmatrix} \quad (39)$$

$$\begin{bmatrix} k_{val}^1 + k_{val}^2 & -k_{val}^2 \\ -k_{val}^2 & k_{val}^2 \end{bmatrix} \begin{bmatrix} U_2 \\ U_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{4} \\ \frac{29}{24} \end{bmatrix} \quad (40)$$

Use Matlab solve the linear system, please see codes in the following.

```
k1 = 4*(log(5.5) - log(5));
k2 = 4*(log(6) - log(5.5));

K = [k1 + k2, -k2; -k2, k2];
F = [1/4; 29/24];

U = K\F;
U2 = U(1);
U3 = U(2);

fprintf('U2 = %.6f, U3 = %.6f\n', U2, U3);
```

Output:

```
c1 = 7.996963, c2 = -0.705297
U2 = 3.825230, U3 = 7.296998
```

Thus, the approximate nodal displacements are:

$$U_1 = 0, \quad U_2 = 3.825230, \quad U_3 = 7.296998 \quad (41)$$

So, the FEM approximate solution is:

$$u(x) = \begin{cases} 7.650460x, & 0 \leq x \leq 0.5 \\ 3.825230 + 6.943536(x - 0.5), & 0.5 < x \leq 1 \end{cases} \quad (42)$$

Use Matlab (codes please see in the following) to plot the exact and the approximate solution using the FEM as shown in Figure 2:

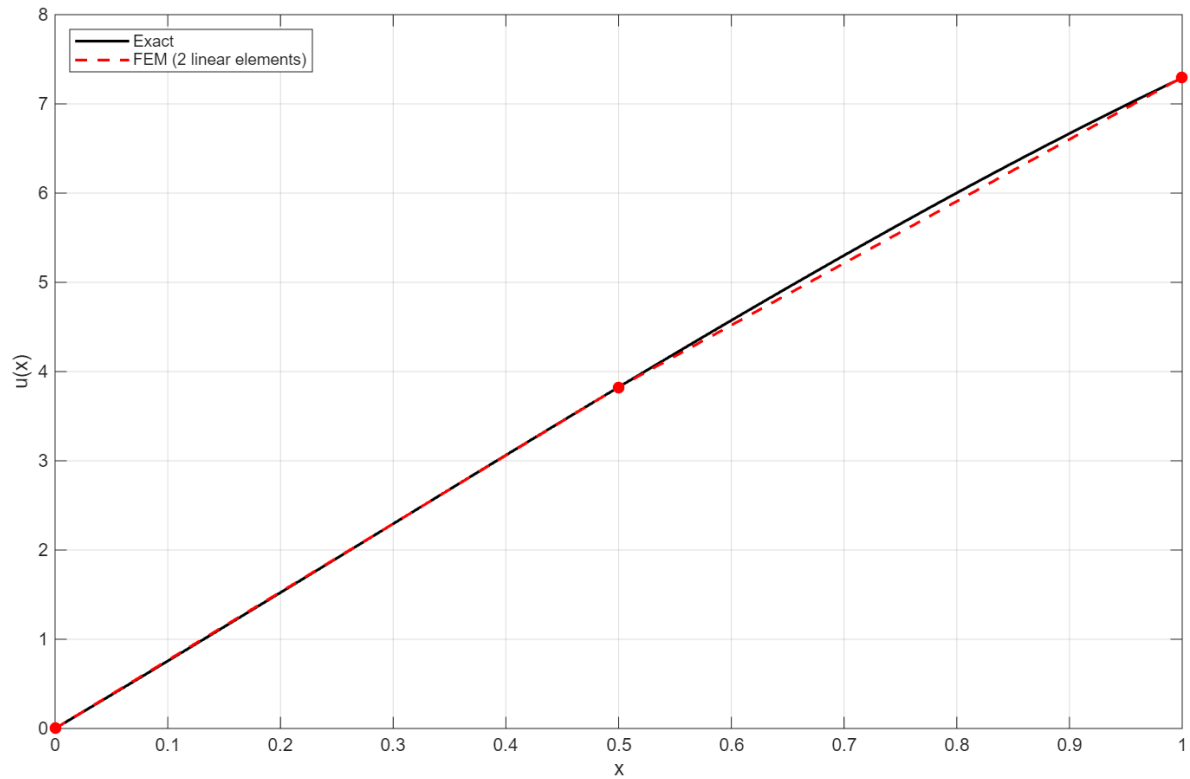


Figure 2: The comparison between exact and FEM solutions

```
% Exact solution
x = linspace(0, 1, 400);
u_exact = -(1/8)*x.^4 - (5/6)*x.^3 + (3/4)*x.^2 + (15/2)*x;

% FEM nodal values (U1 = 0)
k1 = 4*(log(5.5) - log(5));
k2 = 4*(log(6) - log(5.5));
K = [k1 + k2, -k2; -k2, k2];
F = [1/4; 29/24];
U = K\F;
U1 = 0; U2 = U(1); U3 = U(2);

% FEM piecewise linear solution
u_fem = zeros(size(x));
idx1 = x <= 0.5;
```

```

idx2 = x > 0.5;
u_fem(idx1) = U1 + (U2 - U1) * (x(idx1) - 0) / 0.5;
u_fem(idx2) = U2 + (U3 - U2) * (x(idx2) - 0.5) / 0.5;

% Plot
plot(x, u_exact, 'k-', 'LineWidth', 1.5); hold on;
plot(x, u_fem, 'r--', 'LineWidth', 1.5);
plot([0 0.5 1], [U1 U2 U3], 'ro', 'MarkerFaceColor', 'r');
grid on;
xlabel('x'); ylabel('u(x)');
legend('Exact', 'FEM (2 linear elements)', 'Location', 'NorthWest');

```